The Foundation, Explained: $(\mathbb{S}, \tau)$

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Written for a reader with no prior background in this specific mathematics.

$(\mathbb{S}, \tau)$ **is the foundation. A state, a temperature, a specific physical realization — these are scaffolding, kept only until something more fundamental supplies or replaces them. This document says so plainly rather than hedging it.**

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1. The algebra, unconditionally

1.1 What $\mathbb{S}$ is, and how it is built

There is a well-known trick for "doubling" a number system into a bigger one: take *pairs* of numbers from the old system, and define a new multiplication rule that combines them. The oldest example is how the complex numbers are built from the real numbers. A complex number is a pair of real numbers (a, b), usually written $a + bi$, with

$$(a, b)(c, d) = (ac - bd,\ ad + bc).$$

Nothing mysterious happens except a new symbol i appears with $i^2 = -1$; multiplication stays commutative and associative.

Apply the same trick again — pairs of complex numbers — and you get the **quaternions** ($\mathbb{H}$), four-dimensional. Something is lost: quaternion multiplication is no longer commutative.

Double again — pairs of quaternions — and you get the **octonions** ($\mathbb{O}$), eight-dimensional. Something further is lost: octonion multiplication is no longer even associative.

Double once more — pairs of octonions — and you get $\mathbb{S}$, the **sedenions**, sixteen-dimensional. What is lost this time is the property of being a division algebra: two nonzero sedenions can multiply to give exactly zero. This staircase — $\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to \mathbb{S}$ — is the Cayley–Dickson construction, and $\mathbb{S}$ is where this project's formalism begins.

A few properties, stated with equal weight to their failures, since a foundation should own its liabilities as plainly as its virtues:

- **Unital**: there is a direction that leaves everything unchanged when multiplied.
- **Power-associative**: raising a single element to a power is unambiguous.
- **Positive-definite norm**: every nonzero element has a strictly positive "length."
- **Not associative. Not alternative. Has zero divisors** (two nonzero elements whose product is zero) — real features of $\mathbb{S}$, not omissions from a list of virtues.

$\mathbb{S}$'s own symmetries — the ways it can be relabeled without changing anything about its structure — form the group $G_2 \times S_3$.

1.2 τ: splitting $\mathbb{S}$ in two

τ is a specific kind of symmetry: a "flip" with $\tau^2 = 1$, leaving one eight-dimensional half of $\mathbb{S}$ entirely alone and reversing the sign of the other half. The half left alone is exactly the octonions again — call it $\mathbb{O} = \mathrm{Fix}(\tau)$ — and the reversed half is $\mathbb{O}e_8$, everything you get by multiplying an octonion by the specific new direction the last doubling introduced. So $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$: every sedenion splits uniquely into an "even," τ-fixed octonion part and an "odd," τ-flipped part.

Three different, equally good choices of τ exist, related to each other by a further symmetry μ of order three (apply it three times, return to the start). Which one gets used makes no difference to anything built from it — a free, symmetric choice, the same way it makes no difference which of three identical doors you walk through if they all lead to the same room.

1.3 The mediator, explained properly

This is the choice that does the most work in the whole construction, and it deserves a careful explanation rather than a passing mention.

The problem it solves. $\mathrm{Fix}(\tau) = \mathbb{O}$ has a seven-dimensional "imaginary part" — the part left over once you set aside the one direction that behaves like the number 1. Call this $\mathrm{Im}(\mathbb{O})$. On its own, this seven-dimensional space is just a space: seven equivalent directions, nothing marking any one of them as special, nothing organizing them into pairs. But almost everything downstream — a canonical pairing between two sets of three directions, something that behaves like multiplication by i, an actual notion of "position and momentum" — needs *some* single direction singled out first, the way you need to pick an axis before you can define "clockwise."

What gets chosen. A single unit-length direction inside $\mathrm{Im}(\mathbb{O})$, called the mediator. Nothing more than that: one arrow, of length one, pointing somewhere in a seven-dimensional space of otherwise identical directions.

Why this doesn't secretly smuggle in extra information. Here is the fact that makes this a free choice rather than a hidden commitment: G_2, the octonions' own symmetry group, acts *transitively* on the unit sphere inside $\mathrm{Im}(\mathbb{O})$ — meaning any unit direction can be rotated into any other unit direction by some genuine symmetry of the octonions. This is exactly like a perfectly round ball: you can call any point on its surface "the top," and the ball itself has no opinion about which one you picked, because every point looks exactly like every other point until you decide to look at it that way. Whatever gets built from *this* choice of mediator is, structure for structure, identical to what would be built from any other choice — just relabeled. So picking a mediator adds no real information to $(\mathbb{S}, \tau)$; it only fixes *which labels* get used to describe a structure that was already there.

What the mediator actually does, concretely. Call the chosen direction e. Left-multiplication by e — the operation "multiply whatever you're given by e, on the left" — turns out, on the six remaining directions of $\mathrm{Im}(\mathbb{O})$ (everything except e itself), to behave exactly like multiplication by the imaginary unit i: applying it twice brings you back to where you started, with a minus sign. Checked directly: this operation, restricted to those six directions, satisfies $J^2 = -1$ exactly — the defining property of a *complex structure*. That single fact turns six ordinary real directions into three genuinely complex ones, the same way pairing up two real numbers (a, b) into $a + bi$ turns two real dimensions into one complex dimension. This is the origin of the "three pairs" shape that recurs throughout everything built afterward — not assumed, but produced directly by the one thing the mediator supplies: an operation that squares to -1.

From this same e, three more things follow immediately, all standard constructions once you have a compatible pairing of this kind: a symplectic form σ (built from the commutator $[x, y]$, reading off its component along e); the complex structure J just described; and a metric g built compatibly from both. All three together are called the geometric skeleton.

The one thing worth saying plainly: this entire skeleton — σ, J, g, all of it — is fixed the moment $(\mathbb{S}, \tau, \text{mediator})$ is fixed. Nothing further needs to be supplied, and nothing about the skeleton is up for negotiation once those three choices — the algebra, the split, one direction — are made.

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2. What $\mathbb{S}$ already supplies for both sectors, from one multiplication table

2.1 The Clifford relation, native to $\mathbb{S}$

Here is something worth knowing plainly, since it corrects a natural but wrong first guess: the ingredient a fermionic (matter-particle) construction needs does not have to be built abstractly and grafted onto $\mathbb{S}$ from outside. It is already there, in the *odd* half of τ's own split.

For any two directions x, y in $\mathrm{Im}(\mathbb{O})$, look at the corresponding odd-part elements xe_8 and ye_8, and ask what $\mathbb{S}$'s own multiplication does to their anticommutator — the sum of the product taken both ways, $xe_8 \cdot ye_8 + ye_8 \cdot xe_8$. Checked directly: it comes out to exactly $-2\langle x, y\rangle$, twice the ordinary inner product of x and y, times the identity. Two directions at right angles anticommute *exactly* — checked to machine precision, not approximately. A direction paired with itself gives exactly minus twice its own squared length.

This is not a resemblance to the standard relation a fermionic construction needs (the Clifford relation) — it *is* that relation, produced natively by $\mathbb{S}$'s own multiplication table, not imported from outside and bolted onto the odd half afterward.

Meanwhile the *even* half's own multiplication — the ordinary commutator $[x, y]$ for $x, y \in \mathrm{Im}(\mathbb{O})$ — is exactly what §1.3's symplectic form σ was built from. So: one multiplication table, one involution splitting it in two, and each half already supplies, on its own and without further construction, the raw ingredient a different sector needs — the commutator feeding the geometric skeleton on the even side, the anticommutator matching the Clifford relation on the odd side.

2.2 Why a single element can't do the fermionic job alone

A fermionic creation operator needs an unusual property: applying it twice gives exactly zero. Trying to get this directly from a single nonzero element of $\mathbb{S}$, by using that element's own self-product, fails for two independent reasons: every nonzero element has strictly positive length, and squaring can never make a positive length vanish; and a separate, more careful check on the corresponding multiplication operator confirms the same thing at a deeper level — that operator's own eigenvalues turn out to be purely imaginary, about as far from "squares to zero" as a number can get. This doesn't block a fermionic construction — §2.1's relation already supplies what's actually needed — but it closes off the more naive route of building one directly from a single element's self-product.

2.3 Why bosons can't be built from a finite number of fermions

One more thing worth knowing, since it closes off a tempting but impossible shortcut: the canonical position-momentum pair this construction eventually needs cannot be assembled from any *finite* collection of fermionic building blocks, for a reason with nothing to do with this specific project. For any two operators acting on any finite-dimensional space whatsoever, a basic fact of linear algebra says the "trace" of their commutator is always exactly zero. But the position-momentum pair's defining relation demands a commutator whose trace is *not* zero — it has to equal (a multiple of) the total size of the space. Those two facts are only compatible if the space is infinite-dimensional. A finite number of fermionic ingredients, however many, always gives a finite-dimensional space. So this specific shortcut — building the bosonic pair directly out of finitely many fermionic pieces — is not just difficult, it is impossible outright. What survives from §2.1 is a weaker, more honest claim: shared origin in one algebra, not a shared construction.

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3. What is scaffolding, named as such

3.1 What the geometric skeleton doesn't give you

$(\mathbb{S}, \tau, \text{mediator})$ hands you the full geometric skeleton and nothing more. It does not, by itself, tell you what any particular physical situation is actually doing — for that you need a further ingredient, a **state**: a rule assigning an honest expectation value to every measurable quantity, consistent with the skeleton's own rules. Nothing in §1 or §2 forces such a rule to exist, and nothing forces which kind it would be if it did.

3.2 The commitment this project actually makes

This project works with a specific, narrow class of states — regular, quasi-free states satisfying the KMS (thermal-equilibrium) condition. This is a choice, not something $(\mathbb{S}, \tau)$ forces: the space of covariances compatible with the fixed geometric skeleton is far larger than this one family — nothing rules out states that are not in equilibrium at all, or that never reduce to a single number the way this family does. Restricting attention to the equilibrium family is retained here as scaffolding: useful, tractable, and not proven to be necessary.

3.3 β, and exactly what it adds

Given that commitment, the state's covariance at each mode works out to $\nu(\beta) = \frac{1}{2}\coth(\beta k/2)$, where k is the mode's own frequency (already fixed by the skeleton) and β is the one remaining free number — inverse temperature. β does not modify the geometric skeleton; it reweights the one already-fixed geometry, by the same amount at every mode, and no squeezing transformation can undo that reweighting — it is a genuinely physical fact about the state, not a relabeling.

Two boundary values of β behave very differently, and it's worth knowing why. As $\beta \to \infty$ (zero temperature), the covariance approaches the smallest value the geometry allows at all — the vacuum, the one state adding nothing beyond the geometric minimum. This is a limit the family approaches, not a member the family actually contains. As $\beta \to 0$, something different happens: the resulting object isn't even a state of the *same kind* — it has no covariance operator at all, no continuous field structure in the representation it generates. The two edges of the family aren't mirror images of each other; one is a genuine (if unattained) limit of the family's own kind of object, the other falls outside that kind entirely.

3.4 Why different temperatures give different, walled-off worlds

States at genuinely different temperatures, for the same underlying dynamics, turn out to be *disjoint* — not just different, but so different that no ordinary physical process could ever connect one to the other. This is a standard, general fact from the mathematics of infinite quantum systems, not something specific to this construction, and the reason for it is worth having precisely rather than taking on faith.

Each state's own internal sense of time — its modular flow — runs at a rate set by β, relative to one fixed physical dynamics. If two states at different temperatures could be smoothly compared to each other (technically: were "quasi-equivalent"), their two different rates would have to be reconcilable by an adjustment entirely internal to the shared structure — but that would force the underlying dynamics itself to be, in a precise sense, trivial, doing nothing at all. For a genuine, non-trivial physical dynamics, that's a contradiction. So genuinely different temperatures cannot be compared this way; they generate separate, walled-off worlds. The one case where the argument doesn't apply is exactly the degenerate case already excluded in §3.3 — the state where nothing is actually happening at all.

So, within the equilibrium commitment already made, exactly one number remains free: β. And different values of it are not different descriptions of one thing. They are different worlds.

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4. Open

Whether $(\mathbb{S}, \tau)$ could, by some mechanism not yet found, select a specific β rather than leaving it as outside input. One natural candidate has already been checked and ruled out: the most symmetric possible way of reading a number off $(\mathbb{S}, \tau)$'s own structure turns out to give exactly the degenerate, flowless state already excluded in §3.3. No other candidate has been tried.

Whether the equilibrium commitment itself — choosing to work with regular, quasi-free, KMS states in the first place, rather than the much larger space of states nothing currently rules out — will eventually be justified by something more basic, or will simply remain a standing, acknowledged choice.

Whether the placement of §2's two structures — the Clifford relation on τ's odd half, the commutator on its even half — is more than where each one happens to already sit. Whether τ's own grading is doing more structural work than has been shown so far is not yet examined.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.